3. The following staff readings were observed successively with a level 0.875, 1.235, 2.310, 1.385, 2.930, 3.125, 4.125, 0.120, 1.875, 2.030 m.

The first reading taken on BM is 132.135 m. The instrument has been moved after 2nd , 5th and 8th reading. Enter the readings in a field book form and reduce the RLs. Apply usual checks.





Total Station

## 3.4 Advancements in Surveying

### 3.4.1 Introduction

With advancements in technology, new surveying equipments and techniques are developing. Current advancements are making the science of surveying more valuable, accurate and comprehensive than ever.

Few advancements in surveying instruments are discussed below

a. Total station
b. Global Positioning system
c. Geographical Information system

### 3.4.2 Total Station (TS)

It is used for surveying and building construction. It is an electronic transit instrument integrated with electronic distance measurement(EDM) to measure both vertical and horizontal distance and the slope distance from the instrument to a particular point.

Robotic total stations allow the operator to control the instrument from a distance via remote control. This eliminates the need of an assistant staff member as the operator holds the retro-reflector and control the total station from observed point.

DO YOU KNOW?

**Drone survey**

Drone surveys are a faster, safer and more cost-efficient way to survey at height. Sometimes referred to as aerial surveys; UAS (Unmanned Aerial System) surveys, or UAV (Unmanned Aerial Vehicle) surveys, drone surveys are more popular method of surveying.





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